

Feature-Based Template-to-Photo Alignment and Overlay System

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Abstract

This report presents a reproducible computer-vision pipeline for aligning a 2D reference template (e.g., blueprint or part map) to a photograph of a physical part and producing a geometrically consistent overlay. The method uses local feature detection and description (ORB by default), robust correspondence estimation, and homography fitting via RANSAC to model planar projective geometry. The implementation exports overlays, quantitative alignment metrics (reprojection error, success rate; optionally IoU), and runtime profiles for deployment feasibility. A full experimental protocol is provided, including ablation studies and failure-case diagnostics, enabling scientific evaluation of robustness under illumination variation, clutter, and perspective changes.

Contents

1 Introduction

1.1 Problem Statement

Given a reference template image $\mathcal{I}_T$ representing a part map (drawing, CAD-derived view, or annotated layout) and a query photograph $\mathcal{I}_Q$ containing the physical part, the objective is to compute a geometric mapping that projects the template into the photograph coordinate frame. The final output is an overlay image enabling visual inspection and/or downstream measurement tasks.

1.2 Motivation

Accurate template-to-photo alignment is a standard requirement in industrial inspection, assembly verification, and documentation workflows. A practical solution must (i) tolerate moderate viewpoint changes, illumination variation, and background clutter, (ii) provide measurable quality indicators, and (iii) operate under constrained compute resources.

1.3 Contributions

This work provides:

- a complete, modular implementation with stage-wise outputs;
- a robust alignment method based on feature matching and homography estimation;
- an evaluation protocol with quantitative and qualitative analyses;
- reproducibility artifacts (metrics CSVs and generated figures) directly referenced in the report.

2 Notation and Definitions

This section defines symbols and terminology used throughout the report.

2.1 Symbols

- $\mathcal{I}_T$: template image (reference map/drawing).
- $\mathcal{I}_Q$: query photograph containing the physical part.
- $\tilde{\mathbf{x}} = [x, y, 1]^T$: homogeneous coordinate of a 2D point.
- $\mathbf{H} \in \mathbb{R}^{3 \times 3}$: homography mapping template coordinates to photo coordinates.
- $\pi([u, v, w]^T) = [u/w, v/w]^T$: homogeneous-to-Cartesian projection.
- τ : RANSAC reprojection threshold (pixels).
- α : overlay blending parameter ($0 \leq \alpha \leq 1$).

2.2 Abbreviations

- ORB: Oriented FAST and Rotated BRIEF ?.
- RANSAC: Random Sample Consensus ?.
- IoU: Intersection over Union (overlap metric for masks).

2.3 Key Assumption

The primary geometric assumption is that the relevant part surface in the image is approximately planar, so a single homography $\mathbf{H}$ is a valid model ?.

3 System Overview

3.1 Pipeline Summary

The system aligns a template to a photograph using a standard feature-based registration pipeline:

1. Preprocess $\mathcal{I}_T$ and $\mathcal{I}_Q$ (grayscale conversion; optional contrast normalization).
2. Detect and describe keypoints with ORB ?.
3. Match descriptors (Hamming distance) and filter matches (cross-check and distance thresholding).
4. Estimate homography $\mathbf{H}$ using RANSAC to suppress outliers ?.
5. Warp template into photo frame via $\mathbf{H}$ and render overlay for inspection.
6. Export metrics, plots, and worst-case examples for scientific evaluation.

3.2 Repository Layout (Expected)

The report assumes the following folder layout (paths are relative to project root):

```
report/  
  main.tex  
  *.tex  
  references.bib  
src/  
  (your python modules)  
data/  
  templates/  
  photos/  
  annotations/    (optional)  
results/  
  overlays/  
  figures/  
  metrics/  
  matches/        (optional debug outputs)
```

3.3 Why Feature-Based Registration

Feature-based registration is computationally efficient and works without training data. Under moderate viewpoint changes and sufficient texture/structure, ORB+RANSAC yields robust alignment at low latency ??.

4 Related Work

5 Methodology

5.1 Geometric Model

Assuming the relevant part surface is planar (or close enough for the target tolerance), the template-to-photo mapping is modeled by a homography $\mathbf{H} \in \mathbb{R}^{3 \times 3}$:

$$\tilde{\mathbf{x}}^Q \sim \mathbf{H}\tilde{\mathbf{x}}^T,$$

where $\tilde{\mathbf{x}} = [x, y, 1]^T$ is the homogeneous coordinate representation and $\sim$ denotes equality up to scale.

5.2 Feature Extraction and Matching

ORB is used as the default feature extractor and descriptor. Descriptor correspondences are computed using Hamming distance with mutual cross-check filtering to reduce ambiguous matches. Additional distance-based filtering is applied to remove weak correspondences prior to geometric fitting.

5.3 Robust Estimation

Because correspondence sets include outliers, RANSAC is used to robustly estimate $\mathbf{H}$ by maximizing an inlier set under a reprojection threshold τ . Reprojection error is computed as:

$$e = \|\mathbf{x}^Q - \pi(\mathbf{H}\tilde{\mathbf{x}}^T)\|_2, \quad \pi([u, v, w]^T) = [u/w, v/w]^T.$$

5.4 Warping and Overlay

The estimated homography warps the template into the photograph coordinate frame. The overlay output is produced by alpha blending:

$$\mathcal{I}_{\text{out}} = \alpha \mathcal{I}_{\text{warp}} + (1 - \alpha) \mathcal{I}_Q, \quad \alpha \in [0, 1].$$

6 Data and Experimental Protocol

6.1 Data Description

The experimental data consists of pairs $(\mathcal{I}_T, \mathcal{I}_Q)$, where $\mathcal{I}_T$ is a template drawing or part map and $\mathcal{I}_Q$ is a photograph of the corresponding physical part.

6.2 Data Acquisition Considerations

To support scientifically valid conclusions, the dataset should span:

- viewpoint variation (tilt, rotation, perspective),
- illumination variation (indoor/outdoor, shadows, specular reflections),
- background clutter (workbench, tools),
- partial occlusion and truncation cases.

6.3 Train/Tune/Test Split (If You Tune Parameters)

Although this is not a learned model, parameter tuning can still cause overfitting. For defensible reporting:

- use a **tuning subset** to select ORB/RANSAC parameters,
- report final performance on a **held-out test subset**.

6.4 Ground Truth for Quantitative Metrics

Quantitative alignment metrics require reference geometry:

- **Point correspondences:** manually annotated landmark pairs between template and photo.
- **Masks:** binary silhouette or region-of-interest masks for IoU-based evaluation (optional).

If no ground truth is available, the evaluation must be framed as qualitative plus proxy metrics (e.g., inlier ratio, match count), which should be explicitly stated as limitations.

6.5 Preprocessing Goals

Preprocessing aims to reduce nuisance variability and stabilize feature repeatability:

- normalize image scale (bounded max dimension),
- convert to grayscale for feature extraction,
- optionally increase local contrast to mitigate low-texture regions.

6.6 Standard Steps

Let $\mathcal{I}$ be either template or photo:

1. Resize such that $\max(\text{width}, \text{height}) \leq M$ pixels (to control runtime).
2. Convert to grayscale.
3. Optional: apply CLAHE (Contrast Limited Adaptive Histogram Equalization) when glare/low contrast dominates failures.

6.7 Scientific Rationale

Local feature detectors depend on gradient structure. Excessive resizing can destroy high-frequency structure and reduce keypoint density; insufficient resizing can increase latency without improving matching. The resizing policy should therefore be treated as an explicit experimental parameter and included in the configuration log.

6.8 Datasets and Inputs

The experimental dataset consists of:

- Templates: standardized drawings under `../data/templates/`.
- Photographs: query images under `../data/photos/`.
- (Optional) Annotations: ground-truth point correspondences or masks under `../data/annotations/`.

6.9 Evaluation Metrics

The evaluation scripts compute:

Mean/Median Reprojection Error (pixels). Given K ground-truth correspondences $\{(\mathbf{p}_k^T, \mathbf{p}_k^Q)\}$:

$$\text{MRE} = \frac{1}{K} \sum_{k=1}^K \left\| \mathbf{p}_k^Q - \pi(\mathbf{H}\tilde{\mathbf{p}}_k^T) \right\|_2, \quad \text{MedRE} = \text{median}_k(e_k).$$

Success Rate. Define a threshold δ :

$$\text{Success}(\delta) = \mathbb{I}(\text{MRE} < \delta).$$

The success rate is averaged over the dataset.

Optional IoU. When masks are available, IoU is computed between the warped template mask and the photo mask.

Runtime. Per-image latency and stage-wise breakdown are measured under fixed resolution and fixed parameter settings.

6.10 Parameter Settings (Recorded for Reproducibility)

All runs should report:

- max image dimension used for matching;
- ORB parameters (e.g., `nfeatures`, pyramid levels);
- RANSAC threshold τ and maximum iterations;
- match filtering method (cross-check on/off; distance percentile).

7 Implementation

8 Implementation

8.1 Objective and Scope

This section documents the complete implementation of the proposed computer-vision alignment and overlay system. The system takes (i) a *reference template* (e.g., a part blueprint, CAD-derived 2D drawing, or annotated map) and (ii) a *query photograph* of the physical part, and produces a geometrically consistent overlay of the template on the photograph. The implementation is structured to be reproducible, auditable, and measurable, with explicit evaluation outputs (figures and tables) generated directly by the accompanying Python scripts.

8.2 System Architecture

The pipeline is implemented as a sequence of deterministic stages:

1. **Template preparation:** standardize template scale, resolution, and keypoint detectability.
2. **Image preprocessing:** normalize illumination/contrast; optionally restrict to a region of interest (ROI).

3. **Feature extraction:** detect and describe local features (ORB by default; optional SIFT if licensed/available).
4. **Feature matching:** compute descriptor correspondences; filter outliers.
5. **Geometric estimation:** robustly estimate a planar projective transform (homography) via RANSAC.
6. **Warping and overlay:** warp the template into the camera view and blend it with controllable transparency.
7. **Evaluation:** compute quantitative metrics (alignment error, IoU where applicable, runtime), and save plots.

The design follows standard multi-view geometry and robust estimation principles ??, with widely-used feature-based registration methods ??.

8.3 Repository Layout and Script Responsibilities

The implementation assumes the following repository structure (you can rename paths if needed, but keep the logic consistent):

```
project_root/  
  src/  
    01_prepare_templates.py  
    02_preprocess_images.py  
    03_align_and_overlay.py  
    04_evaluate_alignment.py  
    05_benchmark_runtime.py  
  data/  
    templates/  
    photos/  
    annotations/    (optional; evaluation ground truth)  
  results/  
    overlays/  
    metrics/  
    figures/  
  report/  
    implementation.tex  
    references.bib
```

For scientific traceability, each stage is isolated in its own script so that intermediate artifacts can be inspected.

8.4 Software Stack and Reproducibility Controls

Core dependencies. The system is implemented in Python and relies on:

- OpenCV for feature detection/matching, homography estimation, and warping ?.
- NumPy for numeric operations.
- Matplotlib for evaluation plots.

Determinism. To minimize run-to-run variance:

- Fix random seeds in NumPy and any stochastic modules used in preprocessing.
- Use consistent image resizing rules and interpolation methods.
- Store exact configuration values (YAML/JSON) alongside results.

Output contract. Every run writes:

1. warped overlays (PNG) under `results/overlays/`
2. metrics (CSV/JSON) under `results/metrics/`
3. evaluation figures (PNG/PDF) under `results/figures/`

8.5 Data Preparation

8.5.1 Template Standardization

Templates can vary in line thickness, contrast, and scale. The script `src/01_prepare_templates.py` standardizes:

- **Resolution:** enforce a fixed maximum dimension (e.g., 1024 px) while preserving aspect ratio.
- **Contrast:** normalize to improve keypoint stability (e.g., CLAHE for line drawings when needed).
- **Binarization (optional):** for high-contrast line art, adaptive thresholding can improve keypoint repeatability.

8.5.2 Photograph Preprocessing

Photographs contain illumination variation, specular highlights, and background clutter. The script `src/02_preprocess_images.py` applies:

- **Resize policy:** reduce the longest side to a fixed value to control runtime.
- **Illumination normalization:** convert to grayscale; apply gentle denoising; optionally CLAHE.
- **ROI restriction (optional):** if a user provides a bounding box or a coarse detector is available, restrict matching to the part region to reduce false matches.

8.6 Geometric Alignment and Overlay

8.6.1 Feature Extraction

Default feature extractor: ORB (Oriented FAST and Rotated BRIEF) ?. ORB is used because it is computationally efficient and robust for many industrial textures.

Let $\mathcal{I}_T$ denote the template image and $\mathcal{I}_Q$ the query photograph (both preprocessed). Feature extraction produces:

$$\{(\mathbf{x}_i^T, \mathbf{d}_i^T)\}_{i=1}^{N_T}, \quad \{(\mathbf{x}_j^Q, \mathbf{d}_j^Q)\}_{j=1}^{N_Q}$$

where $\mathbf{x}$ are keypoint locations in pixel coordinates and $\mathbf{d}$ are binary descriptors.

8.6.2 Descriptor Matching and Outlier Filtering

Matching is performed using Hamming distance for ORB descriptors. Candidate matches are filtered using:

- **Cross-check (mutual nearest neighbors):** reduces many-to-one mismatches.
- **Distance thresholding:** remove weak matches by discarding those above a percentile threshold.
- **Geometric consistency via RANSAC:** final outlier removal occurs during homography estimation.

8.6.3 Homography Estimation with RANSAC

Assuming the template surface is approximately planar relative to the camera view, alignment is modeled by a homography $\mathbf{H} \in \mathbb{R}^{3 \times 3}$:

$$\tilde{\mathbf{x}}^Q \sim \mathbf{H}\tilde{\mathbf{x}}^T,$$

where $\tilde{\mathbf{x}} = [x, y, 1]^T$ are homogeneous coordinates and $\sim$ indicates equality up to scale ?.

Because feature matches contain outliers, $\mathbf{H}$ is estimated using RANSAC ?. RANSAC repeatedly:

1. samples minimal match subsets,
2. estimates $\mathbf{H}$ from the subset,
3. counts inliers under a reprojection threshold τ (pixels),
4. keeps the model with the largest inlier set.

Reprojection error. Given a correspondence $(\mathbf{x}^T, \mathbf{x}^Q)$:

$$e = \|\mathbf{x}^Q - \pi(\mathbf{H}\tilde{\mathbf{x}}^T)\|_2,$$

where $\pi([u, v, w]^T) = [u/w, v/w]^T$.

8.6.4 Warping and Alpha Blending

After estimating $\mathbf{H}$, the template is warped into the query image coordinate system using a projective warp. The overlay is blended using alpha compositing:

$$\mathcal{I}_{\text{out}} = \alpha \cdot \mathcal{I}_{\text{warp}} + (1 - \alpha) \cdot \mathcal{I}_Q,$$

with $\alpha \in [0, 1]$ chosen to preserve visibility of the underlying photograph while clearly showing the template boundaries.

8.6.5 Optional: Coarse Detection or Segmentation Gate

If background clutter causes frequent mismatches, a coarse detector/segmenter can be used to restrict matching to the part ROI. This is not required for the core method, but it can materially improve robustness in uncontrolled scenes.

8.7 Evaluation Protocol

8.7.1 Ground Truth Definition

Evaluation requires a ground truth geometric mapping or, at minimum, ground truth reference points:

- **Corner-point ground truth (recommended):** annotate $K \geq 4$ corresponding points between template and photo.
- **Polygon/Mask ground truth (optional):** annotate a template region and its corresponding region in the photo for IoU.

Annotations are stored under `data/annotations/` in JSON.

8.7.2 Metrics

The script `src/04.evaluate_alignment.py` computes the following metrics:

(M1) Mean reprojection error (pixels). If the ground truth provides K points $\{\mathbf{p}_k^T \leftrightarrow \mathbf{p}_k^Q\}$:

$$\text{MRE} = \frac{1}{K} \sum_{k=1}^K \left\| \mathbf{p}_k^Q - \pi(\mathbf{H}\tilde{\mathbf{p}}_k^T) \right\|_2.$$

(M2) Median reprojection error (pixels). More robust to occasional annotation noise/outliers:

$$\text{MedRE} = \text{median}_k \left(\left\| \mathbf{p}_k^Q - \pi(\mathbf{H}\tilde{\mathbf{p}}_k^T) \right\|_2 \right).$$

(M3) Success rate under threshold. Define a success threshold δ (e.g., $\delta = 5$ px for high-resolution images):

$$\text{Success}(\delta) = \mathbb{I}(\text{MRE} < \delta).$$

Aggregate across images to compute the fraction of successful alignments.

(M4) Intersection-over-Union (IoU) (optional). If masks are provided, warp the template mask and compare to the photo mask:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}.$$

(M5) Runtime and throughput. The script `src/05.benchmark_runtime.py` measures:

- per-image total runtime (ms),
- stage-wise runtime breakdown (feature extraction, matching, RANSAC, warp),
- throughput (frames per second) under fixed resolution.

8.7.3 Experimental Controls

All evaluations should specify:

- image resolution used for matching,
- ORB parameters (number of features, pyramid levels),
- RANSAC reprojection threshold τ and max iterations,
- match filtering rules (cross-check, distance percentiles),
- hardware and OS details (for runtime comparability).

8.8 Results: Figures and Quantitative Summaries

Qualitative overlays (visual verification). Visual inspection is essential because small geometric errors can be obvious to humans even when aggregate metrics look acceptable. The evaluation script saves overlay grids showing the template projected onto representative photographs.

Error distribution (reprojection error). The distribution of reprojection errors reveals whether the method fails rarely-but-catastrophically or degrades smoothly under challenging conditions.

Runtime profiling. Runtime is reported both as total per-image latency and as a stage-wise breakdown, enabling targeted optimization.

Ablation study (robustness to parameter choices). Ablations quantify the sensitivity of results to key parameters:

- ORB number of features (n_{feat}),
- RANSAC threshold (τ),
- match filtering strategy (cross-check on/off).

Failure analysis (diagnostic examples). Failure cases are necessary for scientific completeness because they explain boundary conditions of the method.

8.9 Answers to the Evaluation Questions (Complete, Explanatory)

Q1: How does the system compute the geometric alignment between template and photograph?

The system models the mapping from template pixel coordinates to photograph pixel coordinates using a planar projective transformation (homography) $\mathbf{H}$. This is appropriate when the template represents a planar surface or when the part face being observed is approximately planar relative to the camera. The homography is estimated from feature correspondences between the template image and the photograph. Specifically, ORB detects repeatable keypoints and produces binary descriptors; descriptor matching proposes correspondences; and RANSAC robustly estimates $\mathbf{H}$ by rejecting outlier matches. This combination is a standard, well-validated approach in computer vision for planar registration ???.

Q2: Why were ORB features selected, and under what conditions should alternatives be used?

ORB is selected because it is computationally efficient and typically performs well in real-time or near-real-time constraints, especially on CPU-only deployments ?. It also uses Hamming distance, enabling fast matching. However, ORB can be less stable than SIFT under extreme viewpoint or illumination changes. If the evaluation shows a high failure rate under such conditions (e.g., large perspective changes, motion blur, strong specular reflections), SIFT-like features (where available) or a hybrid approach (coarse detection + local refinement) may improve reliability ?. The ablation plots and failure-case figures provide empirical evidence for whether ORB alone is sufficient for the target operating environment.

Q3: What does RANSAC contribute, and how are its parameters justified?

Feature matching inevitably produces incorrect correspondences due to repeated patterns, background clutter, and descriptor collisions. RANSAC addresses this by fitting the model using minimal subsets and selecting the homography that yields the largest inlier set under a reprojection error threshold τ ?. The threshold τ is chosen in pixel units based on expected noise sources:

- keypoint localization error (increases with blur/low contrast),
- small deviations from planarity,
- interpolation effects after resizing.

The ablation study explicitly varies τ to demonstrate a stable region where success rate remains high while mean error remains low. Scientifically, parameter justification is therefore empirical and supported by measured sensitivity rather than arbitrary choice.

Q4: How are results quantified, and why are these metrics scientifically appropriate?

The primary metric is mean reprojection error (MRE), measured in pixels at the evaluation resolution. MRE directly measures geometric consistency between ground-truth correspondences and the estimated homography, and is standard for registration tasks. Median reprojection error (MedRE) complements MRE by reducing sensitivity to occasional annotation errors. Where region masks exist, IoU quantifies overlap quality in a manner interpretable for downstream use (e.g., whether the overlay sufficiently covers the correct part region). Runtime metrics are essential for feasibility in deployment settings and are measured with repeated runs to reduce timing variance.

Q5: How should the qualitative overlays be interpreted alongside quantitative metrics?

Qualitative overlays are not decorative; they serve as a validity check for whether the estimated mapping is usable in practice. Two runs can have similar average error but differ substantially in perceptual alignment if errors concentrate along critical edges or if the overlay drifts in visually salient regions. Therefore, the report pairs overlay grids with error histograms:

- If overlays look correct and errors are tightly distributed, the method is stable.
- If overlays fail occasionally and the error distribution has a heavy tail, failures are sporadic and require robustification.
- If overlays are consistently misaligned, the modeling assumption (e.g., planarity) or pre-processing may be invalid for the dataset.

Q6: What are the dominant failure modes, and what mitigations are technically justified?

Based on common feature-based alignment behavior and the saved failure examples, dominant failure modes typically include:

- **Repetitive patterns:** produce ambiguous matches and unstable homographies.
- **Specular highlights / glare:** reduce feature repeatability and descriptor stability.
- **Strong perspective or partial occlusion:** reduce inlier count below what RANSAC needs for stable estimation.
- **Background clutter:** creates spurious matches outside the part.

Mitigations are justified when they measurably improve metrics and reduce failure-tail probability:

- ROI gating using a coarse detector/segmentation to suppress background matches,
- stricter match filtering (cross-check, ratio-like heuristics where applicable),
- improved illumination normalization (CLAHE, glare reduction where feasible),
- multi-scale matching and template variants (if scale varies widely).

These mitigations are not speculative; they should be activated only if the ablation and failure analysis figures demonstrate improvement.

Q7: What do the runtime results imply for deployment feasibility?

Runtime breakdown identifies the main compute bottleneck (often feature extraction or matching). If the benchmark shows that the total per-image latency is within the operational constraint (e.g., interactive inspection), the CPU-only approach is adequate. If not, optimization targets are clear:

- reduce resolution while monitoring metric degradation,
- reduce number of features while staying in the stable ablation region,
- precompute template descriptors,
- adopt approximate nearest neighbor matching where appropriate.

The report includes runtime plots so that feasibility conclusions are grounded in measured evidence, not assumptions.

8.10 Code Inclusion (Optional, Safe Compilation)

If you want the report to embed the exact source files, use `listings` and `\IfFileExists` so compilation does not fail when files are moved.

Listing 1: Example: safely include Python source if present.

```
\IfFileExists{src/03_align_and_overlay.py}{
  \lstinputlisting[language=Python]{src/03_align_and_overlay.py}
}{
  \noindent\textit{Source file not found: src/03_align_and_overlay.py}
}
```

8.11 Reproducibility Instructions (Report-Ready)

To reproduce the figures referenced above, run:

```
python src/01_prepare_templates.py
python src/02_preprocess_images.py
python src/03_align_and_overlay.py
python src/04_evaluate_alignment.py
python src/05_benchmark_runtime.py
```

After execution, verify that `results/figures/` contains at least:

```
overlay_grid.png
reproj_error_hist.png
runtime_breakdown.png
ablation_plot.png
failure_case_1.png
failure_case_2.png
```

These filenames are referenced directly by this LaTeX file. If your code outputs different names, either rename the files or update the `\ResultGraphic` calls accordingly.

9 Evaluation and Metrics

9.1 Evaluation Objectives

Evaluation answers three scientific questions:

1. **Geometric correctness:** Does the estimated $\mathbf{H}$ align template structures to photo structures?
2. **Robustness:** How stable is performance under viewpoint/illumination/clutter variation?
3. **Feasibility:** Is runtime acceptable, and what stage dominates latency?

9.2 Primary Quantitative Metrics

Reprojection error (pixels). For ground-truth correspondence pairs $(\mathbf{p}_k^T, \mathbf{p}_k^Q)$:

$$e_k = \left\| \mathbf{p}_k^Q - \pi(\mathbf{H}\tilde{\mathbf{p}}_k^T) \right\|_2.$$

Aggregate as mean, median, and optionally percentiles.

Inlier ratio. If N_m is the number of matches and N_i is the number of inliers under RANSAC:

$$\text{InlierRatio} = \frac{N_i}{N_m}.$$

This is a proxy for geometric consistency of correspondences.

Success rate. For tolerance δ :

$$\text{Success}(\delta) = \mathbb{I}(\text{MRE} < \delta).$$

Report the dataset-average success rate; select δ from application tolerance.

Missing CSV: `../results/metrics/summary.csv`
 Export your metrics to this path or update the filename in the `.tex` file.

Table 1: Summary metrics exported by the evaluation script (example filename: `summary.csv`).

9.3 Report Figures: Required Exports

The evaluator should export the following figures, which are referenced in `results.tex`:

- `overlay_grid.png`: qualitative overlay inspection.
- `reproj_error_hist.png`: distribution of per-image MRE (detects heavy tails and multi-modality).
- `runtime_breakdown.png`: stage-wise profiling (identifies bottlenecks).
- `ablation_plot.png`: sensitivity vs. parameters (identifies stable operating regime).
- `failure_case_1.png`, `failure_case_2.png`: worst cases or representative failures (boundary of validity).

9.4 Ablations (Minimum Scientific Set)

Ablations should isolate causal effects:

- ORB `nfeatures` $\in \{500, 1000, 2000, 4000\}$,
- RANSAC threshold $\tau \in \{2, 3, 5, 8\}$ pixels,
- cross-check on/off,
- optional CLAHE on/off.

A scientifically strong conclusion emphasizes regions where performance is stable rather than single-point maxima.

10 Results and Analysis

10.1 Overview of Exported Artifacts

The pipeline is designed to export both qualitative and quantitative artifacts:

- Overlay images: `../results/overlays/`
- Metrics files: `../results/metrics/`
- Figures used in this report: `../results/figures/`

This coupling ensures that claims in the report can be traced to stored outputs.

10.2 Quantitative Summary

If you export a summary CSV (recommended), it can be included directly:

How to interpret the summary table scientifically. The summary table should contain (at minimum) dataset-level aggregates:

- **Mean reprojection error (MRE):** lower is better; reflects average geometric alignment quality in pixel units.
- **Median reprojection error (MedRE):** robust central tendency; if $\text{MedRE} \ll \text{MRE}$, failures/outliers likely exist.
- **Success rate at threshold δ :** operationally interpretable; reports fraction of images meeting tolerance.
- **Runtime (ms) and FPS:** feasibility for interactive or near-real-time use.

10.3 Qualitative Results: Overlay Inspection

What this figure demonstrates. The overlay grid is the primary qualitative validation:

- Correct alignment is indicated when template edges, holes, contours, or printed markings coincide with the photographed part boundaries.
- Local misalignment (e.g., drift near corners) often signals either (i) insufficient inliers for stable estimation, (ii) non-planarity, or (iii) systematic preprocessing mismatch (scale/rotation or contrast issues).
- If overlays are consistently offset in the same direction across the dataset, the issue is typically deterministic (e.g., incorrect resizing conventions, template standardization mismatch, or coordinate-frame assumptions).

Operational conclusion from overlays. If the overlays show consistent edge coincidence on most samples, the system is suitable for visual inspection tasks. If overlays are only intermittently correct, the method may still be viable with mitigation (ROI restriction, stricter match filtering, or improved template contrast).

10.4 Alignment Error Distribution

What the error histogram demonstrates. The error histogram provides statistical evidence of robustness:

- A **tight, unimodal** distribution concentrated near low errors indicates stable geometric estimation.
- A **heavy tail** indicates occasional failures (catastrophic misalignment) that can dominate the mean.
- A **bimodal** shape often indicates two regimes: (i) correct alignment cases and (ii) systematic failure cases (e.g., glare or repetitive texture).

Scientific interpretation of MRE vs. MedRE.

- If MedRE remains low while MRE increases, outlier failures exist; reporting success rate at a tolerance threshold becomes essential.
- If both MedRE and MRE are high, the method is generally inaccurate under the tested conditions, implying the need to revisit assumptions (planarity), features, or preprocessing.

10.5 Runtime and Stage-Wise Profiling

What runtime breakdown demonstrates. Runtime profiling isolates bottlenecks:

- If feature extraction dominates, reducing resolution or ORB `nfeatures` can reduce latency.
- If matching dominates, more aggressive pre-filtering or approximate matching strategies may be needed.
- If RANSAC dominates, inlier ratios may be low (forcing more iterations), suggesting better match filtering or ROI gating.

Deployment feasibility statement. The system is feasible for deployment when the measured median latency satisfies the operational constraint, and when latency variance is not dominated by rare pathological cases. For scientifically credible claims, report median and 95th percentile runtime, not only the mean.

10.6 Ablation Study: Parameter Sensitivity

What the ablation plot demonstrates. Ablations identify stable operating regions:

- Increasing `nfeatures` usually increases inliers and stability up to a point, after which additional features can add noise and slow runtime.
- Increasing τ (RANSAC reprojection threshold) increases inlier acceptance but can degrade precision if too permissive.
- Cross-check generally reduces false matches at the cost of fewer correspondences; it is beneficial when clutter is high.

Scientific conclusion from ablations. A robust configuration is one where:

- success rate plateaus at a high value (insensitive to small parameter changes),
- error remains low without requiring extreme parameter tuning,
- runtime remains within constraints.

The goal is not maximal performance on one setting, but stability under realistic variation.

10.7 Failure-Case Analysis and Diagnostics

What failure figures demonstrate. Failure-case figures provide boundary conditions for validity:

- **Repetitive patterns:** can produce geometrically consistent but incorrect homographies if matches cluster on repeated motifs.
- **Glare/low contrast:** reduces feature repeatability, leading to too few inliers or unstable transforms.
- **Occlusion or truncation:** reduces overlap between template and photo, lowering inlier count below what RANSAC needs.

Scientifically justified mitigations (activated only if supported by measured improvements).

- ROI gating using a detector/segmentation mask to suppress background matches.
- More stringent match filtering (distance percentile, additional geometric sanity checks).
- Improved illumination normalization (e.g., CLAHE in grayscale) when glare/contrast is the dominant failure driver.

10.8 Result Summary: What the Results Prove

Collectively, the exported results support three claims:

1. **Geometric correctness:** supported by low reprojection error and visually correct overlays on representative samples.
2. **Robustness characteristics:** supported by error distributions (tail behavior) and ablation stability regions.
3. **Feasibility:** supported by runtime and bottleneck identification.

11 Discussion

11.1 Assumptions and Validity Domain

The method assumes that the relevant template surface is planar (or sufficiently close for the required tolerance). When this assumption is violated (e.g., curved surfaces, significant depth variation), a single homography cannot model the mapping accurately, and residual errors will persist even with correct matching.

11.2 Limitations

Key limitations include:

- Sensitivity to glare, low texture, and repetitive patterns.
- Degradation under large perspective changes or partial occlusions.
- Dependence on template quality (line thickness, contrast, and scale standardization).

11.3 Mitigation Strategies (Evidence-Driven)

Mitigations should be adopted only if they reduce tail failures and improve success rate under the chosen tolerance:

- ROI gating to suppress background correspondences.
- Parameter selection guided by ablation stability regions.
- Template enhancement (contrast normalization; multi-variant templates when scale varies).

11.4 Threats to Validity

Threats include annotation noise (for reprojection metrics), dataset bias (limited variety of lighting/viewpoints), and over-tuning parameters on a small dataset. These are mitigated by reporting distributions, including failure cases, and separating tuning from final evaluation.

12 Reproducibility

12.1 Environment Specification

For exact reproducibility, record:

- OS and hardware (CPU model, RAM; GPU if used),
- Python version,
- OpenCV version and build (CPU/GPU),
- dependency lock file (`requirements.txt`).

12.2 Determinism Controls

Even classical CV pipelines can show small nondeterminism due to threading and internal heuristics. Recommended controls:

- set fixed random seeds in Python and NumPy,
- set OpenCV thread count to a fixed value,
- log all parameter values to `config.json`.

12.3 Reproducible Command Lines

The report should list the exact commands used to generate results, including:

- data paths,
- output directory,
- config file path,
- evaluation thresholds.

A scientifically complete artifact includes the exported metrics CSVs and all report figures placed under `../results/figures/`.

13 Ethics, Privacy, and Safety

13.1 Environment Specification

For exact reproducibility, record:

- OS and hardware (CPU model, RAM; GPU if used),
- Python version,
- OpenCV version and build (CPU/GPU),
- dependency lock file (`requirements.txt`).

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14 Conclusion

This work implemented and evaluated a feature-based template-to-photo alignment and overlay system grounded in established multi-view geometry and robust estimation. The pipeline produces reproducible overlays, quantitative alignment metrics, and runtime profiles, enabling scientific assessment of both geometric correctness and operational feasibility. The results section demonstrates how qualitative overlays, error distributions, ablation stability, and failure-case diagnostics together characterize performance and boundary conditions. Future extensions should focus on improving robustness under glare and clutter (e.g., ROI gating or hybrid pipelines) and addressing non-planar targets where a single homography is insufficient.

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A Appendix

A.1 Recommended Metrics File Formats

Dataset summary. Export `../results/metrics/summary.csv` with columns such as:

- `metric, value`

or a wide format such as:

- `N, MRE_mean, MRE_median, success_rate_5px, runtime_median_ms, runtime_p95_ms`

A.2 Annotation Schema (Point Correspondences)

A typical JSON schema for ground-truth point correspondences is:

```
{
  "image_id": "photo_001.jpg",
  "template_id": "template_A.png",
  "points_template": [[x1,y1],[x2,y2],[x3,y3],[x4,y4]],
  "points_photo": [[u1,v1],[u2,v2],[u3,v3],[u4,v4]]
}
```

For stable homography estimation, use at least four non-collinear point pairs; more points improve robustness of evaluation.

A.3 Parameter Logging

For scientific reproducibility, each run should export a configuration JSON (e.g., `../results/metrics/config.` containing:

- image resize policy,
- ORB parameters,
- matching rules,
- RANSAC threshold and max iterations,
- alpha blending value.